Achouri Youcef Islem

Applied AI and Machine Learning Internship

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Final-year engineering student focused on applied AI, computer vision and NLP. Available January 2027.

EDUCATION

ESI-SBA 2021 - Present

École Nationale Supérieure d'Informatique de Sidi Bel Abbès
State Engineering Degree in Computer Science - fifth year, degree in progress.
Specialty: Artificial Intelligence and Data Structures (AIDS).

SELECTED PROJECTS

Enterprise Operations Intelligence Hub | Python, LangGraph June 2026

- Built the AI-agent orchestration for a third-place NCS Hack 3.0 team prototype, routing operational incidents to specialist agents and reconciling their outputs into an executive brief.
- Integrated Claude-backed structured responses and a deterministic mock mode; kept downstream actions simulated.

Spatial Computer Vision | Python, PyTorch, Babylon.js

- Developed a spatial prototype that segments windows and doors in panoramic imagery using Mask2Former and SAM, then projects those masks onto 3D wall geometry.
- Implemented ray-plane lifting and depth checks for occlusion, with a WebGPU viewer for inspecting reconstructed scenes and lighting.

Arabic Structured Information Extraction | Python, Qwen 2.5, Ollama 2025

- Authored the NLP module for the team's PlanMilt DZ academic GIS project, converting Arabic descriptions into structured scene data.

Neural Network from Scratch | Python, NumPy 2026

- Implemented a two-layer MNIST classifier with vectorized forward propagation, analytical backpropagation and mini-batch gradient descent, without automatic differentiation.

TECHNICAL SKILLS

Machine learning: Python, PyTorch, NumPy, neural networks, model evaluation.
Computer vision: Mask2Former, SAM, panoramic imagery, 2D-to-3D projection.
LLM applications: LangGraph, structured outputs, Qwen 2.5, Ollama, JSON Schema.
Engineering: FastAPI, Docker, PostgreSQL, Git, pytest, Linux.

LEADERSHIP AND COMMUNICATION

Vice President | INGENIUMS Club, ESI-SBA 2023 - 2024

- Co-organized student technology events and programming contests, including Ingeneer, Ingehack and Ingecode.
- Spoke at ESI-SBA Welcome Day and demonstrated technical projects at campus exhibitions.

AWARDS

2nd place, DevFest Hackathon 2023 - GDG Sidi Bel Abbès, Team Infinity.
3rd place, NCS Hack 3.0 - team award, June 2026.

Languages: English and French (strong proficiency).